

Java Basic → String

250 Programming Practice Problems — Foundation Building

Goal: Build strong programming logic before moving to OOP and DSA.

Rule: Solve these yourself first. This PDF intentionally contains no algorithms or solutions.

Recommended order

Finish each section before moving to the next. For every problem, write the code yourself, test normal cases, edge cases, and at least one unusual case where applicable.

1. Java Basics, Variables & Data Types

1. Print your name, age, and department on separate lines.
2. Print the sum of two integers.
3. Read two integers and print their sum, difference, product, and quotient.
4. Read an integer and print its square and cube.
5. Read a number and print its double and triple.
6. Convert a temperature from Celsius to Fahrenheit.
7. Convert a temperature from Fahrenheit to Celsius.
8. Calculate the area of a rectangle from length and width.
9. Calculate the perimeter of a rectangle.
10. Calculate the area and circumference of a circle.
11. Calculate the area of a triangle using base and height.
12. Calculate simple interest using principal, rate, and time.
13. Calculate the average of three numbers.
14. Read a character and print its ASCII value.
15. Read an ASCII value and print the corresponding character.
16. Convert total seconds into hours, minutes, and seconds.
17. Convert total minutes into hours and remaining minutes.
18. Swap two numbers using a third variable.
19. Swap two numbers without using a third variable.
20. Read a two-digit number and print its digits separately.
21. Read a three-digit number and print the hundreds, tens, and units digits.
22. Calculate total bill after applying a percentage discount.
23. Calculate a student's total marks and percentage for five subjects.
24. Convert a given number of days into weeks and remaining days.
25. Calculate the final amount using principal, rate, and time for simple interest.

2. Operators & Expressions

26. Evaluate an arithmetic expression containing +, -, *, /, and %.
27. Find the remainder when one integer is divided by another.
28. Check the result of integer division versus floating-point division.
29. Evaluate an expression using operator precedence.
30. Calculate the last digit of an integer using the modulus operator.
31. Find the sum of the first and last digit of a three-digit number.
32. Calculate the sum of two fractions given numerator and denominator.
33. Calculate the total price of items given quantity and unit price.
34. Calculate a salary after adding a percentage bonus.
35. Calculate a person's age in months and days approximately from years.
36. Convert a three-digit number into its reverse using arithmetic operators.
37. Find the quotient and remainder for two given integers.
38. Calculate the diagonal of a rectangle.
39. Calculate the volume of a cuboid.
40. Calculate the surface area of a cuboid.
41. Calculate the volume of a cylinder.
42. Calculate the total cost after tax and discount.
43. Calculate the percentage increase between an old and new value.
44. Calculate the percentage decrease between an old and new value.
45. Given marks in several subjects, calculate weighted average.

3. if-else & Conditional Logic

46. Check whether a number is positive, negative, or zero.
47. Check whether a number is even or odd.
48. Check whether a person is eligible to vote based on age.
49. Find the larger of two numbers.
50. Find the smaller of two numbers.
51. Find the largest of three numbers.
52. Find the smallest of three numbers.
53. Check whether a year is a leap year.
54. Check whether a character is a vowel or consonant.
55. Check whether a character is uppercase or lowercase.
56. Check whether a character is a digit.
57. Check whether a number is divisible by 5.
58. Check whether a number is divisible by both 3 and 5.
59. Check whether a number is divisible by 2 or 3.

60. Determine pass/fail from marks using a given pass mark.
61. Assign a grade based on percentage ranges.
62. Check whether three sides can form a triangle.
63. Determine whether a triangle is equilateral, isosceles, or scalene.
64. Determine whether a triangle is right-angled using its sides.
65. Calculate electricity bill using slab-based rates.
66. Calculate income tax using different income ranges.
67. Find the middle value among three distinct numbers.
68. Check whether a number is a two-digit number.
69. Check whether a number is a three-digit number.
70. Check whether a given date's month number is valid (1–12).
71. Determine the number of days in a month using month number and leap-year status.
72. Create a simple calculator using two numbers and an operator.
73. Check whether a student gets scholarship based on marks and income.
74. Determine whether a person is eligible for a loan using age and income rules.
75. Determine the quadrant of a point (x, y).

4. switch-case

76. Print the day name for a number from 1 to 7.
77. Print the month name for a number from 1 to 12.
78. Create a calculator using switch and an operator.
79. Print the number of days for a given month number.
80. Convert a numeric menu choice into a selected food item.
81. Create a simple ATM menu with balance, deposit, and withdrawal options.
82. Print a grade description using a grade character.
83. Convert a digit 0–9 into its word using switch.
84. Print the season based on a month number.
85. Create a unit converter menu for km/miles/meters.
86. Create a menu-driven program for rectangle, circle, and triangle area.
87. Create a menu for basic arithmetic operations.
88. Display traffic-light meaning for R, Y, and G.
89. Display a number category for 1, 2, or 3 using switch.
90. Create a simple banking menu with deposit, withdraw, and balance.

5. Loops: for, while, do-while

91. Print numbers from 1 to N.
92. Print numbers from N down to 1.

93. Print all even numbers from 1 to N.
94. Print all odd numbers from 1 to N.
95. Print multiples of 5 up to N.
96. Find the sum of numbers from 1 to N.
97. Find the sum of even numbers from 1 to N.
98. Find the sum of odd numbers from 1 to N.
99. Find the product of numbers from 1 to N.
100. Calculate factorial of N.
101. Print a multiplication table for a given number.
102. Print multiplication tables from 1 to 10.
103. Count the number of digits in an integer.
104. Find the sum of digits of an integer.
105. Find the product of digits of an integer.
106. Reverse an integer.
107. Check whether an integer is a palindrome.
108. Find the first digit of an integer.
109. Find the last digit of an integer.
110. Find the largest digit in an integer.
111. Find the smallest digit in an integer.
112. Count even and odd digits in an integer.
113. Count zeros in an integer.
114. Calculate the digital sum repeatedly until one digit remains.
115. Check whether a number contains a given digit.
116. Print all divisors of a number.
117. Count the divisors of a number.
118. Find the sum of all divisors of a number.
119. Check whether a number is prime.
120. Print all prime numbers from 1 to N.

6. Nested Loops & Pattern Problems

121. Print a square of N stars.
122. Print a rectangle of rows and columns using stars.
123. Print a right triangle of stars.
124. Print an inverted right triangle of stars.
125. Print a right triangle of numbers.
126. Print an inverted number triangle.
127. Print rows containing the same number repeatedly.

- 128. Print a triangle where each row contains 1 to row number.
- 129. Print a triangle with consecutive numbers.
- 130. Print a pyramid of stars.
- 131. Print an inverted pyramid of stars.
- 132. Print a centered number pyramid.
- 133. Print a diamond of stars.
- 134. Print an inverted diamond.
- 135. Print a hollow square.
- 136. Print a hollow rectangle.
- 137. Print a hollow right triangle.
- 138. Print a number square from 1 to N.
- 139. Print a checkerboard pattern using 0 and 1.
- 140. Print alternating 0 and 1 in a triangle.
- 141. Print a multiplication table grid.
- 142. Print Pascal's triangle.
- 143. Print a Floyd's triangle.
- 144. Print an X pattern using stars.
- 145. Print a plus (+) pattern using stars.

7. Methods / Functions

- 146. Create a method to add two integers.
- 147. Create a method to find the maximum of two numbers.
- 148. Create a method to check even/odd.
- 149. Create a method to check prime.
- 150. Create a method to calculate factorial.
- 151. Create a method to calculate power.
- 152. Create a method to reverse a number.
- 153. Create a method to check palindrome number.
- 154. Create a method to find digit sum.
- 155. Create a method to count digits.
- 156. Create a method to calculate GCD.
- 157. Create a method to calculate LCM.
- 158. Create a method to check Armstrong number.
- 159. Create a method to check perfect number.
- 160. Create a method to generate Fibonacci terms.
- 161. Create a method to find the largest digit.
- 162. Create a method to count occurrences of a digit.

- 163. Create a method to calculate rectangle area.
- 164. Create a method to calculate circle area.
- 165. Create a method to convert Celsius to Fahrenheit.
- 166. Create a method that returns the grade from marks.
- 167. Create a method to find the maximum of three numbers.
- 168. Create a method to find the minimum of three numbers.
- 169. Create a method to check leap year.
- 170. Create a menu-driven calculator using separate methods.

8. 1D Array

- 171. Read N integers into an array and print them.
- 172. Print array elements in reverse order.
- 173. Find the sum of all array elements.
- 174. Find the average of array elements.
- 175. Find the maximum element.
- 176. Find the minimum element.
- 177. Count even elements.
- 178. Count odd elements.
- 179. Count positive, negative, and zero elements.
- 180. Search for a given element.
- 181. Count occurrences of a given value.
- 182. Find the index of the first occurrence.
- 183. Find the index of the last occurrence.
- 184. Copy one array into another.
- 185. Print only elements greater than a given value.
- 186. Print only elements smaller than a given value.
- 187. Calculate the sum of elements at even indices.
- 188. Calculate the sum of elements at odd indices.
- 189. Reverse an array in place.
- 190. Check whether an array is sorted ascending.
- 191. Check whether an array is sorted descending.
- 192. Find the second-largest element.
- 193. Find the second-smallest element.
- 194. Find the difference between maximum and minimum.
- 195. Find all duplicate elements.

9. 2D Array / Matrix

- 196. Read and print a matrix.
- 197. Find the sum of all matrix elements.
- 198. Find the maximum element in a matrix.
- 199. Find the minimum element in a matrix.
- 200. Count even and odd elements in a matrix.
- 201. Find the sum of each row.
- 202. Find the sum of each column.
- 203. Find the largest row sum.
- 204. Find the largest column sum.
- 205. Print the main diagonal.
- 206. Print the secondary diagonal.
- 207. Find the sum of the main diagonal.
- 208. Find the sum of the secondary diagonal.
- 209. Find the sum of both diagonals.
- 210. Transpose a matrix.
- 211. Add two matrices.
- 212. Subtract two matrices.
- 213. Multiply a matrix by a scalar.
- 214. Multiply two matrices.
- 215. Check whether a matrix is square.

10. String Fundamentals

- 216. Read a string and print it.
- 217. Print the length of a string.
- 218. Print the first character and last character.
- 219. Print every character on a separate line.
- 220. Count vowels in a string.
- 221. Count consonants in a string.
- 222. Count digits in a string.
- 223. Count spaces in a string.
- 224. Count uppercase letters.
- 225. Count lowercase letters.
- 226. Count special characters.
- 227. Convert a string to uppercase.
- 228. Convert a string to lowercase.
- 229. Toggle the case of every alphabetic character.
- 230. Reverse a string.

- 231. Check whether a string is a palindrome.
- 232. Compare two strings for equality.
- 233. Compare two strings ignoring case.
- 234. Concatenate two strings.
- 235. Count occurrences of a given character.
- 236. Find the first occurrence of a character.
- 237. Find the last occurrence of a character.
- 238. Remove all spaces from a string.
- 239. Replace spaces with underscores.
- 240. Remove a given character from a string.

11. Mixed Foundation Challenges

- 241. Build a number guessing game using loops and conditions.
- 242. Build a menu-driven calculator using methods.
- 243. Build a student marks analyzer using an array.
- 244. Build a simple attendance counter using an array.
- 245. Read N numbers and print statistics: min, max, sum, average.
- 246. Create a frequency table for digits 0–9 in an integer.
- 247. Create a frequency table for characters in a string.
- 248. Find the most common digit in a number.
- 249. Find the most common character in a sentence.
- 250. Check whether a sentence is a palindrome ignoring spaces and case.

Practice Strategy

- Do not look at solutions immediately. Think about input, output, variables, conditions, and loops.
- If stuck, write the steps in plain language first, then convert them to Java.
- Test boundary cases such as 0, 1, negative values, repeated values, empty strings, and small arrays where relevant.
- Re-solve difficult problems after a few days without seeing your old code.
- After these 250, move to Java OOP, Collections, Exception Handling, File I/O, and then DSA.

Topic progression

Basics → Operators → Conditions → switch → Loops → Patterns → Methods → 1D Array → 2D Array → String → Mixed Problems